20	**C**	The heater must heat both the water and the tank. $Q = (mc + C)\Delta T$ $Q = [(10)(4200) + 5000](45 - 25)$ $Q = (42000 + 5000)(20) = 9.4 \times 10^5$ J Using $P = \frac{Q}{t}$, $t = \frac{Q}{P} = \frac{9.4 \times 10^5}{3.0 \times 10^3} = 313$ s
21	**C**	For an ideal gas, $pV \propto T$